

Likelihood-Ranked Selection in Self-Consuming Diffusion: A W_2 Negative Result and a Narrow Minority-Retention Regime

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Abstract

Self-consuming training loops can induce *model collapse*, including loss of rare modes. We study unlabeled likelihood ranking of synthetics in self-consuming DDPM training on imbalanced mixtures. **Primary claim:** selecting by top likelihood does *not* improve sliced W_2 vs. random- k at generation 1 ($\Delta W_2 > 0$ on 23/23 seed-imbalance cells; bootstrap 95% CI of the mean excludes zero; Wilcoxon $p=2.9\times 10^{-5}$, Holm $p=1.4\times 10^{-4}$). **Secondary, post-hoc / narrow-regime observation:** after an original “false stability” hypothesis was falsified, we report that full bottom>rand>top minority retention appears only at moderate imbalance $\rho=5$ ($n=10$, 10/10; exact Wilcoxon $p=0.002$; CIs exclude zero), and already fails for most seeds at $\rho=7$ (2/5) and $\rho=10$ (3/8). Treat that ordering as exploratory (elevated HARK-ing risk), not a pre-registered law. Workshop-scale note on controllable mixtures and tiny Digits checks—*not* ImageNet-scale diffusion. Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an *interventional* selection policy at matched synthetic count and matched real-mix α , with mode-exact metrics.

What is new. A powered falsification of “top- k looks better on aggregate W_2 ,” plus a *narrow-regime* minority-retention observation at moderate imbalance that does *not* generalize to higher ρ or to image-scale models.

Contributions. (1) Matched-budget DDPM protocol with exact mode masses. (2) Primary: top- k does not improve sliced W_2 vs. random- k at generation 1 ($n=23$ cells). (3) Secondary: bottom>rand>top minority retention at $\rho=5$ ($n=10$), unreliable at $\rho=7, 10$. (4) Tiny Digits checks (PCA supports; pixel DDPM / CVAE do not stably replicate the full order).

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng

Table 1: Primary cells (seed $\times \rho$). Seeds 8–9 only at $\rho=5$.

$\rho \backslash$ seed	0	1	2	3	4	5	6	7	8	9
5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
7	✓	✓	✓	✓	✓	—	—	—	—	—
10	✓	✓	✓	✓	✓	✓	✓	✓	—	—

et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Qiao et al. [2026] prove that selection against a *biased local* real-data reference can prune global tail modes and accelerate collapse in siloed settings; their remedy uses collaborative Wasserstein proxy references. We study a complementary failure mode: ranking synthetics by the *generator’s own* denoising-likelihood proxy, with a matched-budget top- k / random- k / bottom- k contrast and minority-mass retention as the primary diversity metric. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

3 Method

3.1 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k :** pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix:** same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace:** pure synthetic.

Thus mix $\neq$ random- k .

3.3 Metrics

Nearest-mean mode masses; minority-mean m ; ratio $r = m^{(1)}/m^{(0)}$ (“retention,” allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests: bootstrap 95% CIs of the mean paired Δ ; Wilcoxon signed-rank with **exact** sign enumeration for $n \leq 16$ (midranks for ties) and a normal approximation for larger n .

4 Experiments

4.1 Primary design ($n=23$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl`. Matrix: $\rho=5$ with $n=10$ seeds (escapes the $n=5$ exact- p floor of 0.0625), intermediate $\rho=7$ ($n=5$), and $\rho=10$ ($n=8$).

Table 2: $\Delta W_2(\text{top} - \text{rand})$ by generation on all 23 cells (positive = top worse). Bootstrap 95% CI; Wilcoxon p ; Holm across five generation-wise tests.

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	p (Holm)
1	+0.089	0.077	1.16	[0.060, 0.121]	23/23	2.9×10^{-5} (1.4×10^{-4})
2	+0.046	0.044	1.04	[0.029, 0.065]	21/23	0.00015 (0.00061)
3	+0.034	0.030	1.14	[0.022, 0.045]	21/23	0.00022 (0.00066)
4	+0.021	0.033	0.62	[0.007, 0.034]	19/23	0.0013 (0.0027)
5	+0.020	0.027	0.73	[0.010, 0.031]	19/23	0.0015 (0.0027)

Table 3: Per- ρ mean r and seed-wise counts (generation 1). Full order = bottom>rand>top.

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	10	0.651	0.366	0.141	10/10	10/10 / 10/10
7	5	0.437	0.266	0.145	2/5	4/5 / 3/5
10	8	0.329	0.247	0.164	3/8	7/8 / 4/8

4.2 Sliced W_2 (primary powered claim)

Generation-wise tests are repeated measures (serially correlated); Holm is conservative. Evidence against top- k improving W_2 is strongest at $g=1$ (CI excludes zero; unanimous sign).

4.3 Retention: stratified secondary analysis

Because strata differ, we do **not** treat a pooled Wilcoxon as the primary claim. We report bootstrap 95% CIs alongside Wilcoxon p . Table 3 gives means and seed-wise order counts; Table 4 gives stratified paired contrasts.

At $\rho=5$ ($n=10$), all seeds agree and exact $p=0.002$ clears $\alpha=0.05$ (Holm-adjusted within the two contrasts: $p=0.0039$). We still frame this as a **narrow-regime** finding: the full order holds on only 2/5 seeds at $\rho=7$ and 3/8 at $\rho=10$. Top-rand CIs exclude zero at $\rho=7$ and $\rho=10$, but bottom-rand’s CI includes zero at $\rho=10$. Pooled $n=23$ tests are secondary (mix heterogeneous strata).

4.4 Real-mix α ablation (illustrative; $n=2$)

At $\alpha=0.5$, seed 0 breaks the full order (bottom $m^{(1)}$ slightly below random) while seed 1 holds—hence 1/2 (means still rank bottom>rand>top). At $\alpha=0.25$, likewise 1/2; at $\alpha=0.75$, top vs. random reverses on average (0/2).

4.5 Transfer checks (directional; $n=3$)

Minimum two-sided exact Wilcoxon p at $n=3$ with all signs agreeing is 0.25; we report Cohen’s d_z against that bound and do not claim powered generalization.

- **8D GMM:** 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 ; order 3/3; top-rand $d_z = -2.56$, bottom-rand $d_z = +1.28$ (exact $p=0.25$).
- **Digits-PCA:** 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order 3/3; top-rand $d_z = -1.72$, bottom-rand $d_z = +2.67$ (exact $p=0.25$).

Table 4: Stratified paired retention contrasts vs. random- k (bootstrap 95% CI of mean Δ ; exact Wilcoxon for $n \leq 16$).

ρ	Contrast	Mean Δ	d_z	95% CI	Exact p
5	top-rand	-0.225	-1.82	$[-0.30, -0.15]$	0.0020
5	bottom-rand	+0.285	+2.08	$[0.21, 0.37]$	0.0020
7	top-rand	-0.121	-1.09	$[-0.21, -0.04]$	0.125
7	bottom-rand	+0.171	+0.92	$[0.03, 0.32]$	0.250
10	top-rand	-0.083	-0.79	$[-0.14, -0.01]$	0.078
10	bottom-rand	+0.082	+0.44	$[-0.03, 0.21]$	0.578



Figure 1: Retention r stratified by ρ (narrow regime at $\rho=5$).

- **Digits pixel-space DDPM (8×8 MLP denoiser):** order 1/3 at $\rho=5$; means 1.25 / 0.88 / 1.23; bottom>rand on 3/3 seeds ($d_z=+1.27$) but top vs. random is inconsistent (exact signs 1/3 for top<rand). A small image-diffusion check, not ImageNet-scale validation; full ordering not replicated.
- **Digits-CVAE (inconclusive):** pixel-space conditional VAE did not reproduce a stable bottom>rand>top order.

4.6 Selection-pool size k_{frac} (illustrative; $n=2$)

Top- k remains worst on means; at 0.75, bottom vs. random flips (0/2 full order).

Table 5: Mean r by α (seeds $\{0, 1\}$, $\rho=5$, default $k_{\text{frac}}=0.5$). “Order holds” = bottom>rand>top on $g=1$ minority mass. The $\alpha=0.5$ row is the *same cell* as $k_{\text{frac}}=0.5$ in Table 6.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	1/2
0.75	1.34	0.77	0.94	0/2

Table 6: k_{frac} ablation (seeds $\{0, 1\}$, $\rho=5$, $\alpha=0.5$). $0.25 \equiv 0.5$ when n_{syn} binds. The $k_{\text{frac}}=0.5$ row duplicates Table 5’s $\alpha=0.5$ row.

k_{frac}	bottom	rand	top	Order holds
0.25	0.62	0.49	0.24	1/2
0.50	0.62	0.49	0.24	1/2
0.75	0.48	0.61	0.22	0/2

5 Analysis

Top- k retains high-likelihood (typically majority) synthetics. Aggregate sliced W_2 is a poor monitor of minority survival: top- k is consistently *worse* on W_2 at $g=1$ across the full matrix, while minority retention shows a *narrow* bottom>rand>top regime only at $\rho=5$. That regime already fails for most seeds at $\rho=7$ and $\rho=10$.

6 Limitations

Scale. All primary results are on 2D/8D GMMs and tiny Digits (8×8) checks. We have *not* demonstrated these effects on any real image-scale diffusion model (e.g. CIFAR/ImageNet UNet or DiT); Digits pixel-space DDPM does not stably replicate the full retention order (1/3). **Scope.** Workshop / short empirical note—not a definitive scaling result. α and k_{frac} ablations use two seeds; their default rows are the same experimental cell. **Multiplicity:** Holm within the W_2 generation family and within the two $\rho=5$ retention contrasts; other tests descriptive. Complementary to MAD/LSF/Feng and siloed-selection theory [Qiao et al., 2026].

Researcher degrees of freedom and post-hoc revision (important). This note is the survivor of a long, documented exploratory search across many unrelated directions before self-consuming diffusion (archived under **docs/** in the public repository). Within this topic, the *original* working hypothesis was a “false stability” dissociation: top- k would *improve* sliced W_2 while secretly accelerating minority collapse. The pilot **falsified** that dissociation (top- k worsened W_2), after which the claim was revised to the present imbalance-stratified retention ordering plus an explicit W_2 negative result. Readers should therefore treat the bottom>rand>top ordering as a **post-hoc** / **exploratory** finding with elevated HARKing risk—even though $\rho=5$ now uses $n=10$ seeds and exact $p=0.002$ —and should **not** read it as a pre-registered general law. The W_2 negative result is comparatively more trustworthy: it was the observation that killed the original hypothesis, then held on the expanded $n=23$ cell matrix with unanimous sign at $g=1$. We mitigate overclaiming by making W_2 primary in the abstract, stratifying retention by ρ , demoting pooled tests, and leading with bootstrap CIs.

Note on double-blind review. The public GitHub URL reveals author identity. For double-blind venues, provide an anonymous code mirror; for arXiv, the named repository is appropriate.

7 Conclusion

Top- k likelihood selection does not improve sliced W_2 vs. random- k at generation 1 in our matrix. A bottom>rand>top minority-retention order appears in a **narrow** moderate-imbalance regime ($\rho=5$) and should not be stated as a general law. Preferring “most likely” synthetics is not a free lunch on either aggregate W_2 or rare-mode survival when minorities remain measurable.

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A Reproducibility

python scripts/reproduce_w2.py; stats tables via experiments/scripts/make_submission_stats.py.
Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.